

Two Targeted Follow-Up Experiments

Batch A: building ADA/XRP confidence past the 30-sample threshold · Batch B: is 5-minute dead, or pattern-restricted?

Generated 2026-06-21 · Built directly on the existing auto-simulator and edge registry — no infrastructure changes

This addendum runs two follow-up experiments flagged as open questions in the original Model Validation Report: (1) most ADA/XRP pattern combinations were still in "learning" status after the held-out test, and (2) the conclusion that 5-minute is "structurally unsuited" was based on testing all pattern types together — it was never isolated against pattern type specifically. Both experiments use the existing `lib/auto-simulator.ts` and `db.edgeStats` unchanged; the only code addition was a `simConfigOverrides` passthrough field (plus a UI textarea to set it) so a controlled run can override `allowedPatterns` / `minRR` / `minConfidence` without touching `simulator.ts`, `pattern-engine.ts`, or `edge-database.ts`.

Batch A — ADA / XRP / 4h Confidence Building

Config: ADA + XRP only · 15m/1h/4h (5m excluded) · random date windows over the last 10 years · 40 scheduled iterations, target PF set to 3.0 so the batch ran to completion rather than early-stopping.

Sample counts before vs. after — 3 of 38 combinations were already "active" (30+ samples); 7 are now

PATTERN	SYMBOL	TF	BEFORE	AFTER	STATUS	BEFORE WR	AFTER WR	NOTE
Head & Shoulders	XRP	4h	1	1	learning	0%	0%	
Head & Shoulders	ADA	1h	0	22	learning	—	64%	new, close to threshold
Head & Shoulders	XRP	1h	0	26	learning	—	65%	new, close to threshold
Double Bottom	ADA	15m	1	14	learning	0%	79%	SAMPLE-SIZE CORRECTION
Hammer	XRP	15m	6	17	learning	83%	47%	SAMPLE-SIZE CORRECTION
Bearish Engulfing	ADA	15m	0	32	ACTIVE	—	44%	crossed 30
Double Top	XRP	15m	16	37	ACTIVE	31%	27%	crossed 30, stayed weak
Head & Shoulders	ADA	15m	4	46	ACTIVE	100%	57%	SAMPLE-SIZE CORRECTION — crossed 30
Bullish Engulfing	XRP	15m	26	58	ACTIVE	58%	62%	crossed 30, confirmed promising

Bearish Engulfing	XRP	15m	41	87	ACTIVE (already)	54%	54%	stable
Double Bottom	XRP	15m	41	107	ACTIVE (already)	37%	33%	stable, weak
Head & Shoulders	XRP	15m	84	167	ACTIVE (already)	65%	63%	stable, strong

Full 38-combination table available in `training-exports/baseline-adaxrp-agg.json` and `lean-post-batchA.json` . Remaining combinations (mostly ADA 1h/4h and XRP 4h) are still under 30 samples — fewer than 30 candlestick patterns even formed on those symbol/timeframe pairs across the random windows drawn.

The most informative outcome of this batch is exactly the kind the 30-sample threshold exists to catch: Head & Shoulders on ADA 15m looked like a perfect pattern at 100% win rate — on just 4 trades. With 46 trades, it settled at a real 57% win rate. Same story for Double Bottom on ADA 15m (0%→79% on 1→14 trades) and Hammer on XRP 15m (83%→47% on 6→17 trades). None of these are failures of the system; they're the system correctly refusing to trust a result until there's enough of it, then revising as data arrives — which is the entire point of the sample-size gate in `lib/simulator.ts` 's scoring tiers.

4h-specific breakdown — first dedicated data, and it's weak

TIMEFRAME	RUNS	AVG RETURN	AVG PF	AVG WR	PROFITABLE	0-TRADE RUNS
15m	18	+2.2%	1.76	34%	8/18	7
1h	10	+0.2%	1.54	38%	4/10	4
4h	4	-0.2%	0.20	8%	0/4	3

4h is not yet a usable data point. Only 4 runs were scheduled and 3 of those returned zero trades — a 30-90 day random window on a 4-hour chart is only ~180-540 candles, often too few for a pattern to even form and clear the confidence floor. The one run that did trade dragged the average to a 0.20 profit factor. This is "not enough data," not "4h doesn't work" — it needs a dedicated batch with longer windows (6+ months per draw) before it can be judged at all, the same way 15m/1h needed volume before Batch 2 of the original report could draw a real conclusion.

0-trade runs — same root cause as the original held-out test

14 of 32 recorded runs (44%) returned zero trades — 8 of those on ADA, 6 on XRP, concentrated on pre-2020 windows (thin early listing history) and short windows generally. This matches the report's existing finding precisely: short random windows on less liquid pairs legitimately produce no signal. It is not a new problem introduced by this batch.

Batch B — 5-Minute: Structural Failure, or Pattern-Type Artifact?

Hypothesis under test: the original report's "5m is structurally unsuited" conclusion (0/9 profitable, -26.2% avg return, PF 0.73) was measured across *all* pattern types together. This batch isolates pattern type as a variable: same symbols (BTC/ETH/SOL), same 9 fixed quarters, 5m only — but restricted to the six candlestick-reversal patterns (Morning Star, Bullish/Bearish Engulfing, Hammer, Shooting Star, Head & Shoulders) and excluding every multi-bar chart formation (Double Top/Bottom, triangles, flags, wedges, rectangles).

CONDITION	RUNS	AVG RETURN	AVG PF	AVG WR	PROFITABLE
Original 5m, all patterns (prior report)	9	-26.2%	0.73	40%	0/9 (0%)
Variant 1 — candlestick-only, same params (1% risk, R:R 1.5, conf. 65)	18	+0.6%	0.95	51%	7/18 (39%)
Variant 2 — candlestick-only, looser params (R:R 1.2, conf. 60)	16*	-0.7%	0.99	51%	9/16 (56%)

* Variant 2 was scheduled for 18 runs; the verification script's polling window closed mid-run-18, so 16 runs are reported rather than 18. Re-running for the missing 2 would only add precision, not change the conclusion below.

Verdict: supports hypothesis 2 — 5m has a narrow, pattern-restricted edge, it is not categorically dead. Restricting to candlestick patterns moved the profitable-run rate from a clean 0% to 39-56%, and profit factor from 0.73 to 0.95-0.99. That is a real, meaningful shift, not noise. But it is not yet a tradeable result either: profit factor is still under 1.0 in both variants, and average return is roughly flat (one variant slightly positive, one slightly negative). The honest read is "no longer structurally dead, not yet structurally sound" — worth a dedicated follow-up on the one pattern that's actually carrying it (next paragraph), not a green light to re-enable 5m broadly.

Per-pattern breakdown — Hammer is doing almost all of the work in Variant 1

PATTERN	V1 WIN-MENTIONS	V1 LOSS-MENTIONS	V2 WIN-MENTIONS	V2 LOSS-MENTIONS
Hammer	6	1	5	7
Bearish Engulfing	3	2	2	0
Morning Star	2	4	4	4
Bullish Engulfing	2	4	1	2
Head & Shoulders	1	0	0	0
Shooting Star	0	4	3	6

In Variant 1, Hammer was mentioned as a strong performer in 6 runs against only 1 weak mention — by far the cleanest ratio of any pattern, and the clear driver of that variant's positive average return. Bearish Engulfing also held up reasonably (3 good vs 2 weak in V1, 2 good vs 0 weak in V2). **Shooting Star was the worst performer in both variants** (0 good/4 weak in V1, 3 good/6 weak in V2) — worse, proportionally, than some of the chart patterns it was meant to outperform. Head & Shoulders barely fired at all on 5m (1-2 mentions total across both variants) — plausible, since it's structurally a multi-bar pattern despite being grouped with "candlestick" signals for this experiment, and a 5-minute chart may simply not give it room to form as often as 15m/1h do.

Variant 1 vs Variant 2 is genuinely ambiguous, not a clean result either way. Looser entry parameters (R:R 1.2, confidence 60) raised the profitable-run rate from 39% to 56%, but average return went from +0.6% to -0.7% and profit factor was statistically indistinguishable (0.95 vs 0.99). More runs are profitable, but the average outcome is

flat-to-slightly-worse — consistent with looser filters letting in more small wins and more small losses without changing the underlying edge. This is not strong enough evidence to recommend looser parameters for candlestick patterns generally; flagging it as open rather than forcing a conclusion.

Edge-registry separation — confirmed, not assumed

`db.edgeStats` keys every record by pattern × symbol × timeframe × regime × session. Batch B's Hammer/Shooting Star/etc. trades on 5m accumulate into the same per-pattern buckets that already existed from the original all-patterns 5m history (e.g. Hammer-BTCUSDT-5m), which is correct — it's the same pattern being traded under the same conditions, regardless of which experiment produced the trade. What does *not* happen, by construction of the keying, is any mixing between different patterns: Double Top's existing 5m history is untouched by this batch, since no Double Top trades were possible under the allowlist. Checked directly against the post-batch export rather than assumed from the code.

Summary

- **Batch A worked as intended:** active (30+ sample) ADA/XRP combinations went from 3 to 7, and surfaced two clear sample-size corrections (Head & Shoulders ADA 15m, 100%→57% WR) that validate why the 30-trade gate exists in the first place. 4h remains unjudged — needs longer windows, not more runs of the same length.
- **Batch B narrowed, but did not close, the 5m question:** candlestick-only restriction lifted profit factor from 0.73 to ~0.95-0.99 and the profitable-run rate from 0% to 39-56% — a real improvement, not yet a tradeable edge. Hammer is the standout pattern worth a dedicated, larger follow-up batch; Shooting Star should probably be dropped from any future 5m candlestick test, since it underperformed even the chart patterns it was meant to be tested against.
- Neither batch required changes to `pattern-engine.ts`, `simulator.ts` core scoring, or `edge-database.ts`. The only code change was an additive `simConfigOverrides` passthrough in `lib/auto-simulator.ts` and a matching advanced-options textarea in the auto-sim page, both optional and backward compatible.

Report generated from live Dexie/IndexedDB simulation data pulled directly from the auto-simulator's edge-stats and auto-sim-run tables. All figures are taken from real backtest runs against historical Binance market data.